Advanced Theory — Module 03: Perception — Study Questions

1. Define the particular partition. What are external, sensory, active, and internal states?
2. State the Markov blanket condition precisely. What does conditional independence mean here?
3. How does the Markov blanket create the "inside" and "outside" of a system?
4. State the Free Energy Lemma. Under what conditions do internal state dynamics approximate free energy minimization?
5. What does it mean to say internal states "parameterize" a distribution over external states?
6. Why doesn't the system need to "know" it's performing inference?
7. What is a nonequilibrium steady state? How does it differ from thermodynamic equilibrium?
8. State the Helmholtz decomposition of dynamical flow. What are the dissipative and solenoidal components?
9. Why is solenoidal flow important? What happens without it?
10. How does the dissipative component perform gradient descent on log-density?
11. What is the path integral formulation of Active Inference? How does it relate to the principle of least action?
12. How does the Lagrangian in the path integral relate to free energy?
13. What is "dual aspect monism" in Bayesian mechanics?
14. What is gauge symmetry in the mapping from internal states to beliefs? Why does it matter?
15. How does Bayesian mechanics extend classical mechanics?
16. What role does the particular partition play in defining the boundary between agent and environment?
17. Can the Markov blanket be applied to multi-scale systems? How?
18. What is the relationship between the NESS density and the generative model?
19. How does noise (random fluctuation w) contribute to the dynamics?
20. Critically evaluate: Is the Free Energy Principle unfalsifiable? What would disprove it?